F FWallSim · Firewall Course

CLASS 18

The branch office is cut off

Troubleshooting: IPsec

Advanced | ~15 minutes | FWallSim v1.1.0 (FortiOS 7.4 style)

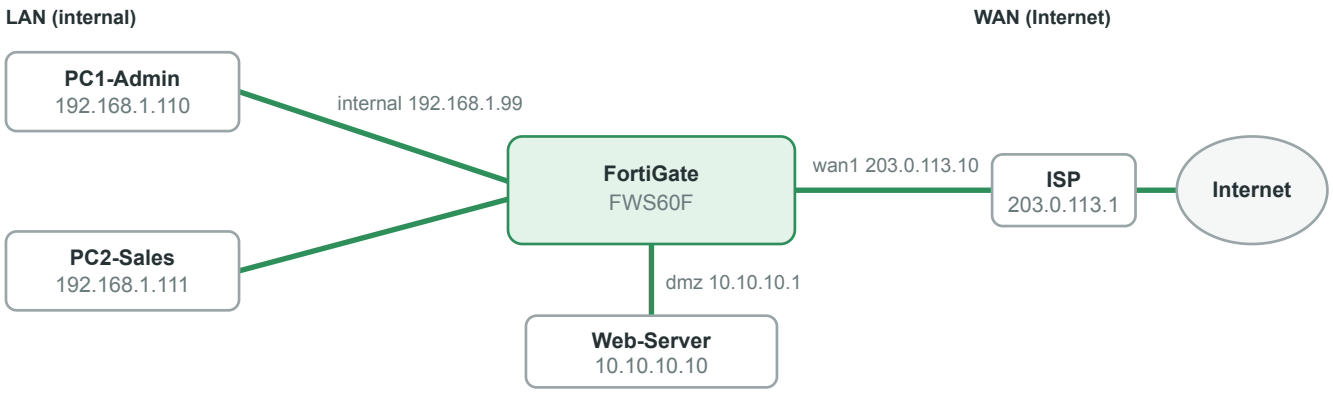
YOU WILL LEARN

- Troubleshooting: IPsec

BEFORE YOU START

Finish Class 17 first.

The lab network



The goal

THE IDEA

Since a change last night, HQ and the branch cannot reach each other. Two things are wrong.

How to start

1. Open FWallSim and log in.
2. Click **Labs** in the header.
3. Find **The branch office is cut off** and click **Start**.
4. The Lab Guide panel opens. Each task below gets a green tick as soon as it works.

Tasks

1. **PC1 can ping Branch-PC**

   Start with VPN > IPsec Tunnels: is the tunnel up? After the fix, send PC1-Admin → 10.20.0.10, Ping (Traffic Generator quick test "PC1 pings Branch-PC").

2. **Branch-PC can RDP to PC1**

   Test the reverse direction too: send Branch-PC → 192.168.1.110, RDP (TCP/3389).

Hints

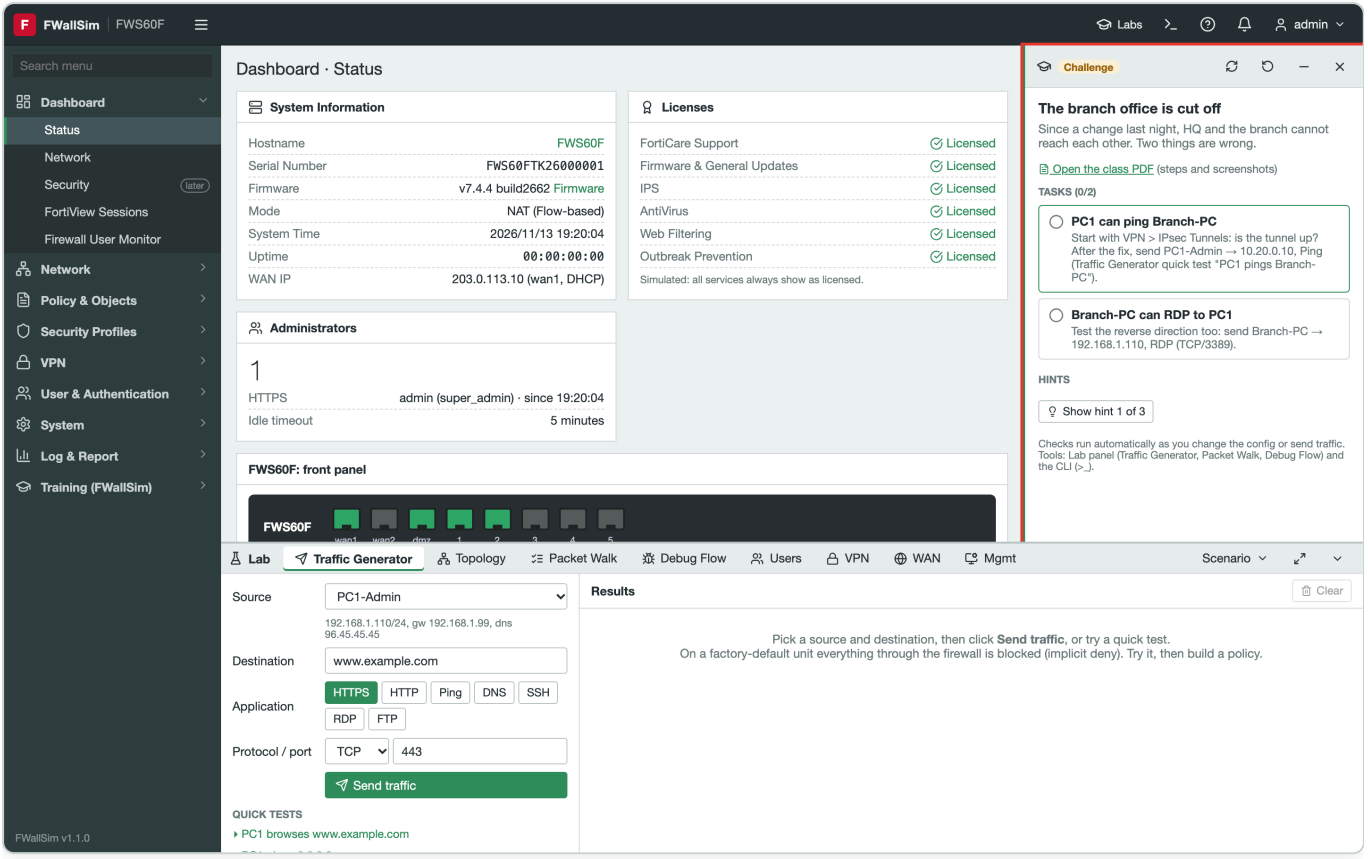
1. IPsec Tunnels shows a reason for the tunnel being down. In the CLI, try "diagnose debug application ike -1" and "diagnose debug enable".
2. Compare the pre-shared key with Lab > VPN.
3. Once the tunnel is up, check the routing table for 10.20.0.0/24.

Solution

1. Edit tunnel to-branch: pre-shared key Branch-PSK-2026.
2. Network > Static Routes > Create New: 10.20.0.0/24, Interface to-branch.
3. Test both directions.

Screenshots

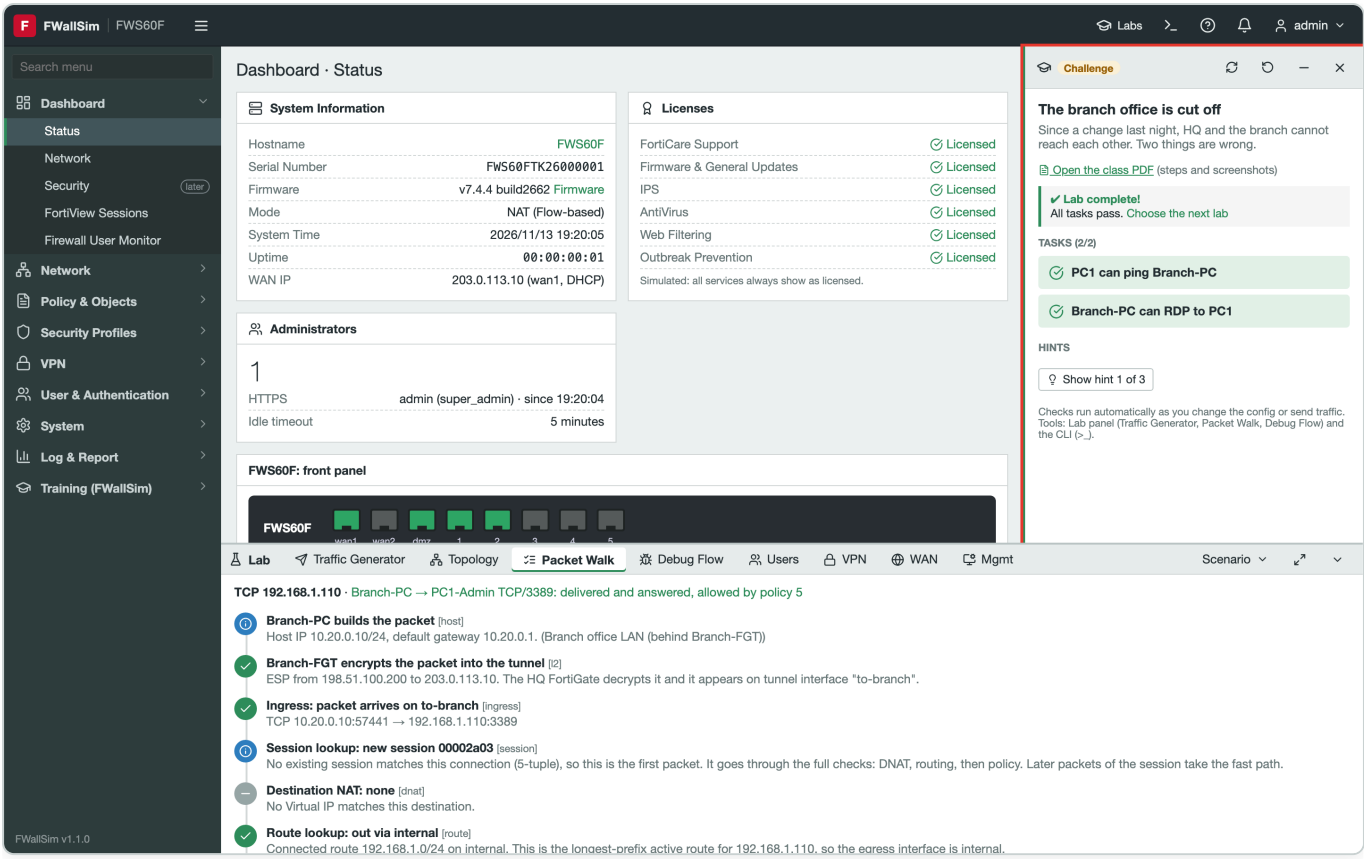
Numbered red markers show where to click or what to look at.



Lab Started



Packet Walk Before



Lab Solved

IPsec Tunnels

+ Create New | Edit | Delete | Bring Down to-branch | Search

Tunnel	Interface Binding	Remote Gateway	IKE	Status	Phase 2 Selectors	Reason
Site to Site - FortiGate (1)						
to-branch	wan1	198.51.100.200	v2	Up	▲ to-branch: 192.168.1.0/24 ↔ 10.20.0.0/24	Established

The branch side is fixed: see **Lab > VPN** for the settings you must match. When a tunnel is down, the Reason column (and `diagnose debug application ike -1` in the CLI) tells you why.

IPsec Tunnels

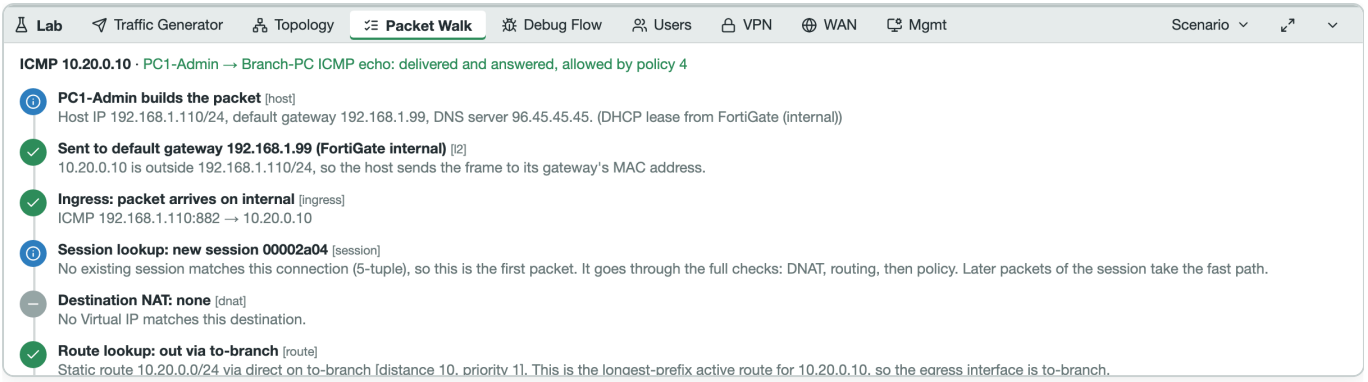
Static Routes

+ Create New | Edit | Clone | Delete | Search

Destination	Gateway IP	Interface	Distance	Priority	Status	Comme
10.20.0.0/24		Blackhole	254	1	Inactive: higher distance than another route to the same prefix	
10.20.0.0/24	0.0.0.0	to-branch	10	1	Active	

The default route learned by DHCP on wan1 does not appear here. See it under Dashboard > Network > Routing. Lower distance wins for the same prefix; for equal distance, lower priority wins.

Static Routes



Packet Walk After

Next: Class 19: The backup line never takes over